STEFAN

SYMPATHETIC SITAR

A halo of strings that rings along with everything you play.



Sympathetic-resonance pedal for MOD Desktop and MOD Dwarf

USER MANUAL · NOVEMBER 2026 EDITION

WHAT SYMPATHETIC SITAR DOES

Underneath the playing strings of a real sitar lies a second set of strings the player never touches: the **taraf**, or sympathetic strings. They are tuned to the notes of the raga, and they ring on their own whenever the music passes over their pitch — a silvery halo that shadows every phrase.

Sympathetic Sitar gives that halo to **any instrument**. Send a guitar, a synth, a voice — anything — through it, and a bank of up to **48 tuned strings** rings along with what you play. You choose the tuning: the strings follow the **scale and key you pick**, from familiar Western scales to Turkish makams and Hindustani ragas with their authentic in-between notes.

Play a note the strings know, and it blooms. Play a chord, and the whole bank shimmers back at you, slowly fading like a piano with the sustain pedal down. You are not adding an effect on top of your sound — you are playing a roomful of strings with it.

> **Good to know:** the strings only ring at *their own* pitches. Put the pedal *after* the effects you want the strings to hear (drive, wah …) and *before* big ambient effects like reverb and delay, so the halo washes into the room with everything else.

QUICK START

1. **Add Sympathetic Sitar to your pedalboard.** In the MOD web interface, find it under the brand **Stefan** and drag it into your signal chain.
2. **Pick your key.** Set **ROOT** to the key of your song and leave **SCALE** on Major (or pick the scale you are playing in).
3. **Press TEST.** The pedal plucks each string in turn, low to high, so you can hear exactly what the halo is tuned to. Press again to stop.
4. **Play.** Notes that belong to the scale make the strings bloom and ring after you.
5. **Balance it.** Turn **MIX** up for more halo, and **DECAY** up for a longer ring.

> **The one trick to learn:** match **ROOT** (and SCALE) to the music you are playing. Strings tuned to the wrong key will still ring — at the wrong notes. If the halo ever sounds sour, the tuning is the first thing to check.

THE CONTROLS



The **lotus** in the top-left corner switches the pedal on and off: open means on, a folded bud means bypassed. It follows a footswitch or MIDI control assigned to the pedal's On/Off too. The pencil (✎) in the top-right corner opens the scale editor.

Selectors & buttons

CONTROL	WHAT IT DOES
1 **SCALE**	Which notes the strings are tuned to. **24 scales**: fourteen Western (major, minors, modes, pentatonics, blues …), four Turkish makams (Rast, Uşşak, Hicaz, Saba) and six Hindustani ragas (Yaman, Bhairav, Bhairavi, Todi, Marwa, Malkauns) — the makams and ragas include their authentic in-between pitches that a piano cannot play — plus any you build yourself (see *Make your own scale*).
2 **ROOT**	The key the scale is built on: **C through B**. Match it to your song.
3 **STEREO**	How the strings spread across the stereo image. **Mono** keeps them centred; **Linear** lays them low-to-left, high-to-right; **Wide** alternates strings far left and far right. The Narrow variants are the same layouts at half width.
4 **TEST**	Plucks each active string in turn, **low to high**, a couple of seconds each — hear the whole tuning without playing a note. Stops by itself at the last string, or press again to stop early.
5 **STRUM** ↑ ↓	Strums every active string once, the way a sitarist brushes the tarafs: ↑ sweeps **low to high**, ↓ **high to low**. Like a hand entering the strings, it eases in: the first strings are plucked more gently. It plays on top of whatever you are doing and sounds even with nothing plugged in. Press again mid-strum to start over. On the MOD Dwarf, give each arrow its own footswitch and alternate strokes.

Knobs

6 GATE

Keeps **hiss and hum** from ringing the strings between phrases.

- **Zero:** everything excites the strings, noise included
- **Low (default):** tames a quiet noise floor
- **High:** only confident playing gets through — ringing tails are never cut off, only new excitement

7 SENS

How **easily the strings ring**.

- **Low:** resonance builds slowly — a faint, distant halo
- **Max (default):** strings answer your full playing strength

8 STRINGS

How many strings ring, counted **from the root upward** (1–48).

- **Few:** a focused chord of resonance
- **13 (default):** almost two octaves of scale
- **Many:** a dense four-octave curtain

9 OCT

The **octave** the string bank starts in (2–6).

- **2:** deep drone territory
- **3 (default):** sits with guitar and voice
- **5–6:** strings float above your notes and only their harmonics ring — pure sparkle

10 DECAY

How long the strings **ring on** after you stop.

- **Low:** a dry, percussive pluck
- **Noon (default):** a graceful one-breath fade
- **High:** minutes-long, nearly a drone

11 BLOOM

Lets ringing strings **excite each other** through a shared bridge, the way a real sitar's tarafs do.

- **Off (default):** each string rings alone
- **High:** tails swell and blossom with extra shimmer

12 JAWARI

The **buzzing bridge** of a real sitar. Each ringing string sings its own vowel as it fades — a bright “ee” at the pluck gliding down to a dark “oo”.

- **Off (default):** clean, bell-like ringing
- **High:** the full sitar yowl on every string

13 LEAD

The same **vowel sweep on your own notes** — every note you play glides from “ee” to “oo” as it decays, like a sitar's main string. The strings still hear your clean sound.

- **Off (default):** your instrument, untouched
- **High:** a sitar-like voice on everything you play

14 MIX

The balance between **your dry sound** and **the strings**. Loudness stays even across the whole sweep.

- **Left:** dry only
- **Right:** strings only — play the resonance itself

15 LEVEL

Output volume trim, **±12 dB**, after the mix.

- Use it to match the pedal's on/off loudness by ear

16 SWEEP

How long a **strum** takes from its first string to its last (20 ms–2 s).

- **Short:** a quick brush, notes blur into one shimmer
- **250 ms (default):** a relaxed stroke
- **Long:** a slow harp-like run, every note distinct

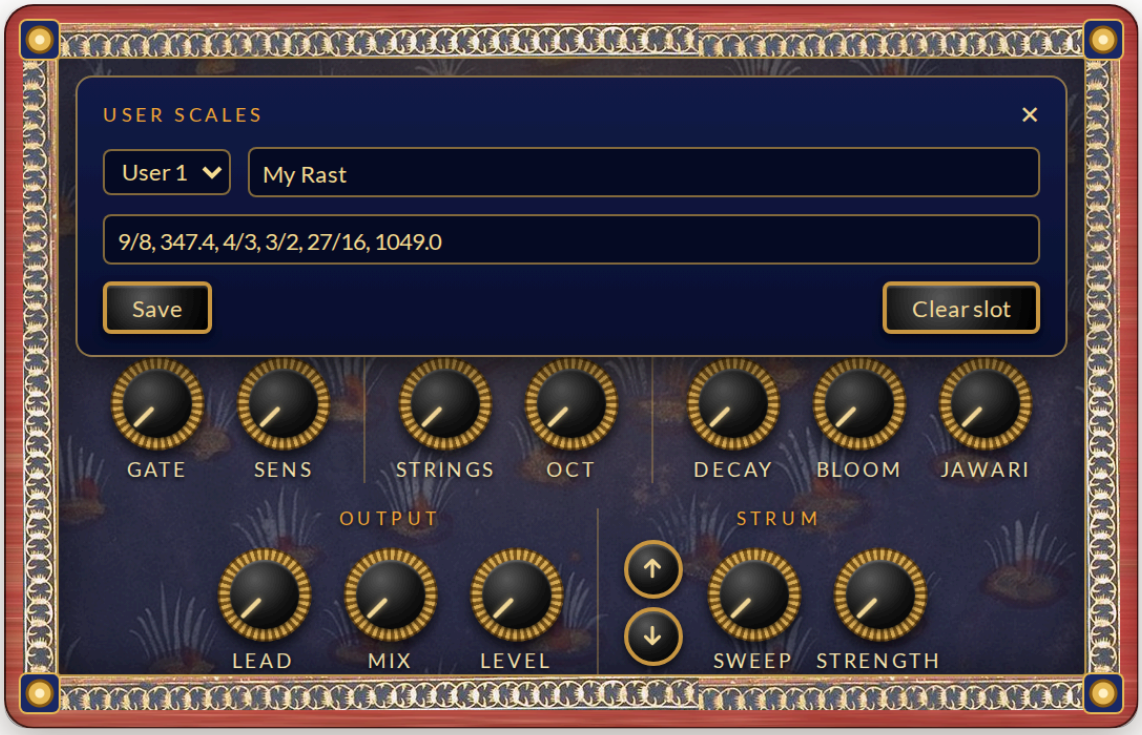
17 STRENGTH

How **hard** the strum plucks. Every string gets the same force, and each strum is plucked afresh, so no two sound quite alike.

- **Low (default):** a soft brush under your playing
- **High:** a bold stroke — with JAWARI up, a harder strum also buzzes more

MAKE YOUR OWN SCALE

The 24 built-in scales cover a lot of ground — but you can also define **your own**, with any intervals you like, including the microtonal pitches between the piano keys. Custom scales are saved with your pedalboard.



1. Click the **pencil** (✎) in the pedal's **top-right corner** to open the **User Scales** editor.
2. Pick a **slot** (User 1–8) and type a **name**.
3. Enter the **intervals** separated by commas. Each is either a **ratio** such as `9/8` or `3/2`, or a value in **cents** — a number with a decimal point, such as `347.4`. The root (`1/1`) is added for you, so just list the notes above it.
4. Press **Save**. Your scale now appears in the **SCALE** menu under **User**; select it like any other scale, and press **TEST** to hear the tuning.

> **Ratios or cents?** Ratios are exact just-intonation, ideal for harmonic tunings and for matching a raga or makam by ear. Cents match how most tuning tables and theory tools write intervals — you can mix both in the same scale.

To change a scale, reopen the editor, choose its slot, edit and **Save** again; **Clear slot** removes it. Custom scales live in the pedalboard — save a MOD preset to carry one between projects.

RECIPES

Knob positions are given like a clock face: fully left is 7 o'clock, the middle is noon, fully right is 5 o'clock.

Classic taraf halo

SCALE Major · **ROOT** your key · **STRINGS** noon · **OCT** 3 · **DECAY** noon · **BLOOM** off · **JAWARI** 9 o'clock · **MIX** 10 o'clock

The real-sitar experience: every phrase you play trails a silver shadow. Fingerpicked guitar practically turns into a different instrument.

Shimmer ceiling

OCT 5 · **DECAY** 1 o'clock · **BLOOM** noon · **JAWARI** off · **MIX** noon · **STEREO** Wide

The strings sit far above your notes, so only their harmonics ring — an airy sparkle floating over everything, close to a shimmer reverb but tuned to your song.

Tanpura drone bed

SCALE Raga Yaman · **STRINGS** 9 o'clock · **DECAY** 3 o'clock · **BLOOM** 1 o'clock · **JAWARI** noon · **MIX** 2 o'clock

A few strings that barely stop ringing. Play the root and fifth a couple of times to charge them up, then improvise over the drone with your hands free.

Microtonal colour

SCALE Hicaz or Saba · **ROOT** to taste · **DECAY** noon · **JAWARI** 10 o'clock · **MIX** 11 o'clock

The makams tune strings to pitches between the piano keys. Press TEST and listen — then play simple lines and let the halo answer with intervals your frets don't have.

Subtle studio thickener

MIX 9 o'clock · **DECAY** 10 o'clock · **GATE** 2–3 · **STEREO** Wide Narrow · **LEVEL** to match

Barely-there resonance that widens and sweetens a part without announcing itself. If you can point at the effect, turn MIX down a hair more.

TIPS

When in doubt, press TEST

After every scale or root change, TEST plays you the exact tuning of the halo. It is the fastest way to learn what the makams and ragas sound like, too.

Sour halo? Check the key

Strings ring at their own pitches no matter what you play. If the resonance clashes, ROOT or SCALE doesn't match the song. Chromatic scale follows anything — at the cost of focus.

Raise GATE until the silence is silent

Amp hiss and hum will happily ring the strings all day. Turn GATE up until nothing sounds when you are not playing — tails still ring out in full.

Roll your own tuning

Beyond the 24 presets, the pencil (✎) editor in the top-right corner builds **custom microtonal scales** from ratios or cents — see *Make your own scale*. They save with the pedalboard.

Bass in, mud out? Raise OCT

With a bass-heavy source, the low strings can ring into mud. OCT 4 or 5 moves the whole bank above the muck, keeping just the sheen.

More strings, same loudness

The pedal balances its own volume as STRINGS changes — many strings each ring slightly softer, so sweeping the knob never blows up the mix.

INSTALLING SYMPATHETIC SITAR

Download the build for your platform from the GitHub Releases page — each release ships a MOD Desktop bundle (`sitar-vX.Y.Z-linux-x86_64.tar.gz`) and a MOD Dwarf bundle (`sitar-vX.Y.Z-dwarf-aarch64.tar.gz`).

MOD Desktop (Linux)

1. Extract the archive: `tar xzf sitar-vX.Y.Z-linux-x86_64.tar.gz`
2. Move the extracted `sitar.lv2` folder into `~/Documents/MOD Desktop/lv2/`
3. Restart MOD Desktop. The pedal appears in the plugin library under the brand **Stefan** as **Sympathetic Sitar**.

MOD Dwarf

1. Connect the Dwarf over USB and extract the Dwarf archive on your computer.
2. Copy the bundle onto the unit:

   ```
   scp -O -r sitar.lv2 root@192.168.51.1:/root/.lv2/
   ```

3. Restart the audio services (or simply reboot the Dwarf):

   ```
   ssh root@192.168.51.1 'systemctl restart jack2 mod-ui'
   ```

4. Hard-refresh the MOD web interface in your browser (**Ctrl-Shift-R**) and drag the pedal onto a pedalboard.

Building from source (any LV2 host, macOS/other platforms) is covered in the project README.

QUESTIONS & ANSWERS

The strings ring notes that clash with my song.

The halo is tuned to SCALE + ROOT, not to what you play. Set ROOT to your song's key and pick the scale you are actually playing in — or choose Chromatic, which contains every note.

I hear hardly any resonance.

Check that MIX is up and SENS is high. If GATE is set high, quiet playing may not be getting through to the strings. And if OCT is 5–6 with a low-pitched instrument, only faint harmonics reach the strings — bring OCT down.

The ringing never seems to stop.

That is DECAY in its drone range, and BLOOM extends tails further still. Back DECAY down toward noon for song-friendly fades.

Can I make my own scale?

Yes — click the pencil (✎) in the pedal's top-right corner to open the User Scales editor, enter the intervals as ratios (like `3/2`) or cents (like `347.4`), and Save. It joins the SCALE menu under “User” and is stored with your pedalboard. See *Make your own scale*.

Can I strum the strings without playing?

Yes — press STRUM ↑ or ↓. The strum ignores GATE and SENS, so it sounds with your instrument silent; SWEEP sets its speed and STRENGTH how hard it plucks.

Does JAWARI make it louder?

No — it adds the buzz without changing level. If you want the halo louder, that is MIX; if you want the whole pedal louder, that is LEVEL.

SPECIFICATIONS

FORMAT	LV2 audio plugin, mono in → stereo out
STRINGS	Up to 48 tuned resonators spanning four octaves above the root
TUNINGS	24 built-in scales (14 Western, 4 Turkish makams, 6 Hindustani ragas — just intonation, with authentic neutral intervals) plus up to 8 user-defined microtonal scales
ROOT RANGE	C–B pitch classes, octaves 2–6 (C2 ≈ 65 Hz to top strings at C8 ≈ 4186 Hz)
PLATFORMS	MOD Desktop (Linux x86_64), MOD Dwarf (aarch64), any LV2 host
LICENSE	ISC (free and open source)

Sympathetic Sitar is free software built with the DISTRHO Plugin Framework. The pedal's artwork uses details of ragamala paintings from The Metropolitan Museum of Art's Open Access collection (public domain): *Solanki Raga* (ca. 1590), *Dipak Raga* (1630–40) and *Ragini Bhairavi* (18th century). Fonts: Rakkas and Lato (SIL Open Font License). Manual edition November 2026 — for the plugin version it shipped with; see the GitHub release page for the latest.